



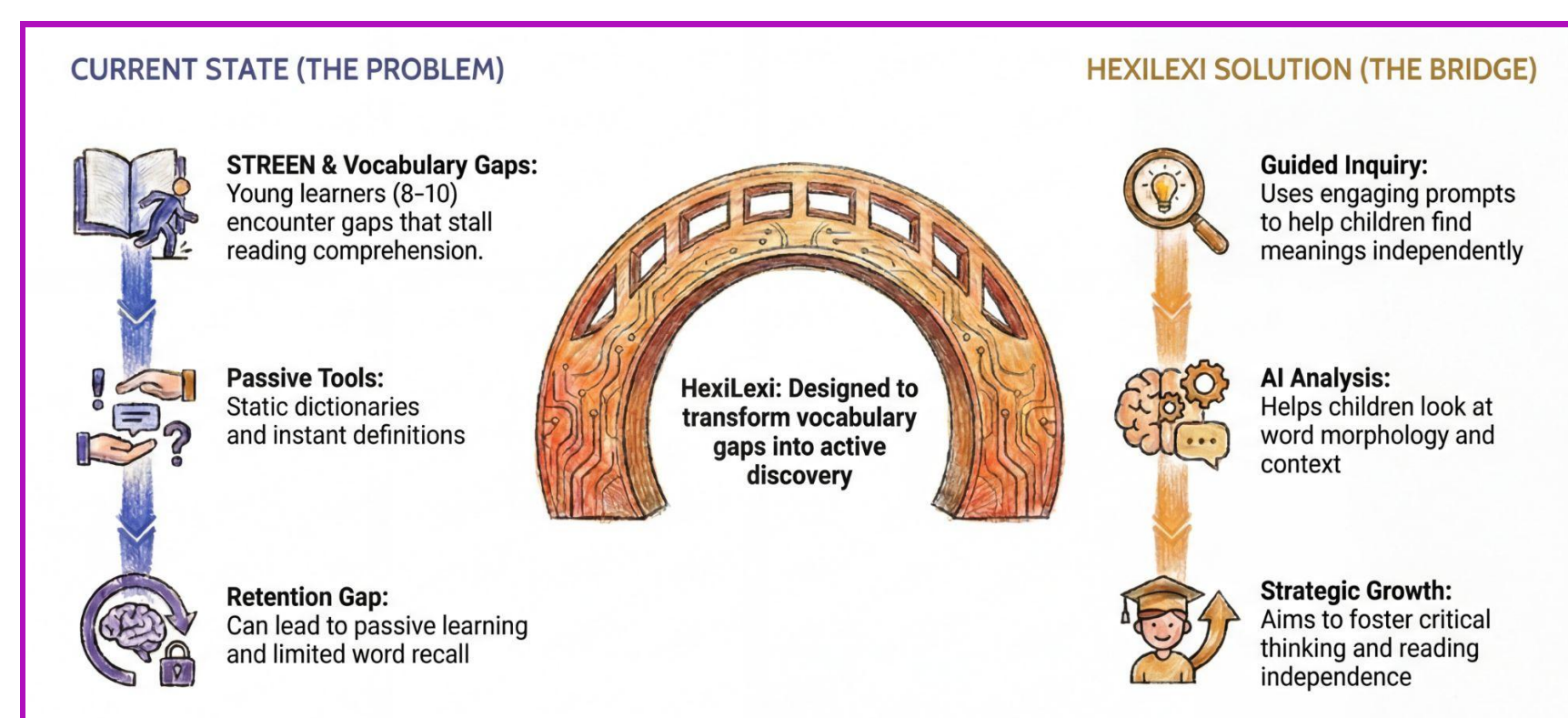
HexiLexi - AI Word Detective

LLM-driven mentor, replacing dictionaries for an interactive learning experience.

Problem Statement

Unfamiliar vocabulary often disrupts reading comprehension & meaning making for young learners. In digital environments like the **STREEN** platform (Ribeiro et al., 2017), vocabulary gaps frequently impede understanding. Traditional supports, such as dictionaries or human intervention, often promote passive learning & result in limited long-term retention (Ebbers & Denton, 2008; Loftus-Rattan et al., 2016).

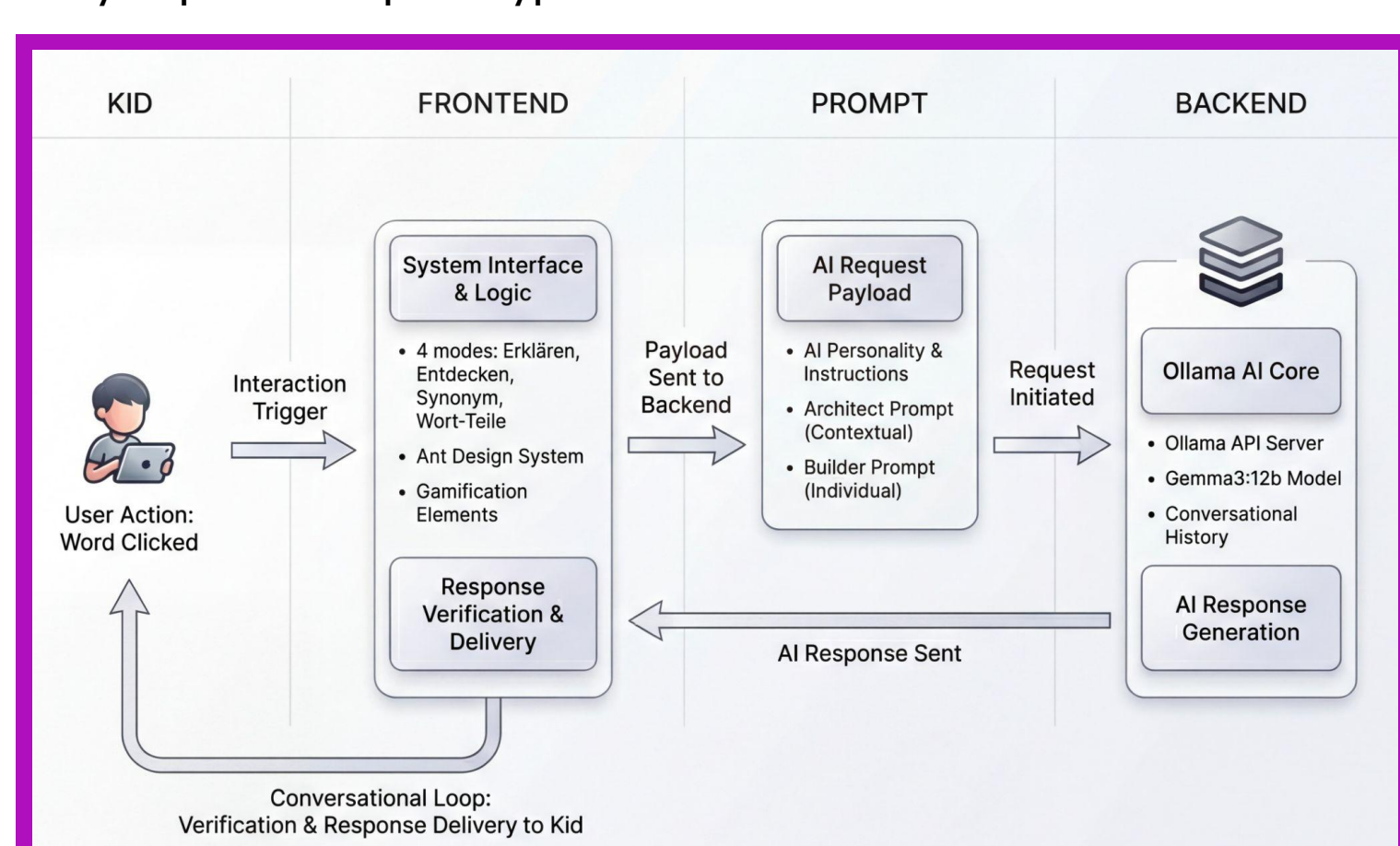
To address this, we developed **HexiLexi**, an AI-based platform within STREEN designed for active meaning construction through active discovery. Rather than providing immediate definitions, HexiLexi uses AI-driven, inquiry-based prompts to guide children through word **morphology & contextual** clues of the story. This transforms uncertainty into strategic exploration, fostering independent learning. By leveraging **word consciousness** (Scott & Nagy, 2004), HexiLexi aims to promote the critical thinking necessary for lifelong linguistic independence.



Methodology & Implementation

HexiLexi is a React-based component designed for seamless integration into digital reading environments. It utilizes a Gemma 3:12B language model to power four distinct learning modes, ranging from direct explanation to scaffolded discovery. To ensure pedagogical accuracy, the system employs a two-step AI process: an initial analysis of the word's context followed by the generation of tailored, inquiry-based prompts.

To identify design challenges, user-centered sessions were conducted with 13 third to fourth-grade students (8:3 girls/boys) in primary school reading laboratories. Participants engaged with specifically designed short stories containing challenging vocabulary to encourage system use. To assess functionality and design, children rated their experience on a 7-point Likert scale using the UX Kids Questionnaire (UXKQ, Wöbbekind et al., 2021). These descriptive statistics, gathered across multiple sessions, were used to iteratively improve the prototype.



Iterative Development through UC Sessions

Session 1: Observation & Strategy Exploration

- ★ We observed natural peer explanations using a word-description game (Tabu), we investigated the role of non-verbal cues, such as emotional & situation facial expressions, since they are important for comprehension (Adler, 2011).
- ★ To understand the suitability of explanatory strategies for helping children derive word meanings more efficiently, we applied clarification questions, guided self-correction, & positive feedback (Watzlawick).
- ★ We presented the initial HexiLexi prototype for feedback through questionnaires.

Session 2: Prototype Testing & Feedback Gathering

- ★ Children used HexiLexi to explore challenging vocabulary in a new story; to assess their understanding children wrote the meanings of the words on a worksheet.
- ★ We collected qualitative feedback through a poster activity (what worked well, what didn't & what can be improved) & quantitative data using UXKQ questionnaires.

Session 3: Broader Evaluation & Motivation

- ★ We introduced a third story with diverse target words to evaluate HexiLexi's descriptive range & response to challenging environments.
- ★ We reapplied Session 1 explanatory strategies & boosted engagement through a team-based word explanation challenge.

Findings & Feedback

Quantitative Findings:

- ★ **System Performance:** Children consistently reported the system worked well ($M=6.23$, $SD=0.83$) & responded quickly ($M=5.23$, $SD=1.30$). Ease of use was also rated high ($M=4.92$, $SD=1.71$).
- ★ **Variation in Experience:** While comprehensibility ($M=5.31$, $SD=2.02$) & motivation ($M=4.54$, $SD=2.22$) were rated positively on average, high standard deviations indicate significant individual differences in these areas.
- ★ **Visual Design:** The platform was generally viewed as neat ($M=6.38$, $SD=0.870$) & cheerful ($M=6.00$, $SD=1.15$). However, diverse preferences existed regarding colorfulness ($M=5.31$, $SD=1.80$) & perceived entertainment value ($M=4.31$, $SD=1.97$).

Qualitative Feedback:

- ★ **Strengths:** Participants generally praised the visual design (*font, appearance*) & found the system fun & motivating with understandable tips.
- ★ **Areas for Improvement:** Some children found vocabulary explanations difficult, repetitive, or disliked the points system. Suggestions included simpler explanations, clearer pronunciation, & adding images to support usability.

Implications & Next Steps

HexiLexi shows promise for supporting vocabulary learning through active meaning construction. Quantitative & qualitative results indicate positive user experiences & increased engagement, aligning with research on the role of word consciousness in language learning (Scott & Nagy, 2004). While children valued the interactive prompts, suggesting potential motivational benefits, conclusions regarding motivational effects are limited by the lack of a validated measure. Additionally, technical issues during Session 3 prevented pre-post data comparison. Future work requires stable deployments & longer-term interactions to systematically examine HexiLexi's impact on vocabulary acquisition, motivation, & reading comprehension.

Project Management:

Prof. Dr. Nele Wild-Wall
Prof. Dr. Christian Ressel
M. Sc. Pedro Ribeiro
Dr. Kyra Kannen

Students:

Emily Heye
Leon Deda
Muhammad Sufyan
Noah Schönwetter
Rashmika-Nawanjana
Taha Asif

Special Thanks:

Huzefa Mustafa
Mirko Musumeci
Children of the Ernst -
Reuter - Grundschule &
Leselab

References:

- ★ Ribeiro et al. (2017): (10.55612/s-5002-032-006)
- ★ Wöbbekind et al. (2021): (<https://doi.org/10.1145/3473856.3473875>)
- ★ Adler (2011): (<https://doi.org/10.17433/978-3-17-022910-5>)
- ★ Ericksen et al. (2024): (<https://scholarworks.gvsu.edu/gradprojects/465>)

Technologies:

react.dev
ant.design
axios-http.com
ollama.com
nodejs.org
jamovi.org



Scan for a quick video
demonstration!